

Using GPU causes Data to not be found

<https://github.com/Qiskit/qiskit-aer/issues/662>

ROW 9

Using GPU causes Data to not be found #662

New issue

Closed

#717



shreychowdhary opened on Mar 15, 2020 · edited by shreychowdhary

Edits ...

Informations

- Qiskit Aer version: 0.4.1
- Python version: 3.6
- Operating system: ubuntu

What is the current behavior?

It throws the following exceptions after running the code for a while:

```
Traceback (most recent call last):
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/result/result.py", line 265, in _get_experiment_data
    exp = next(result for result in self.results
StopIteration

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "nvqe.py", line 211, in <module>
    fidelity,energy,iterations = vqe.run()
  File "nvqe.py", line 162, in run
    ret = optimizer.optimize(num_vars=self.num_params, objective_function=self.objective_func, initial_point=self.initial_point, **self.options)
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/aqua/components/optimizers/spsa.py", line 139, in optimize
    **self.options)
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/aqua/components/optimizers/spsa.py", line 190, in _optimize
    cost_plus = obj_fun(theta_plus)
  File "nvqe.py", line 99, in objective_func
    total = self.objective_fun_rec(params,self.N,None)
  File "nvqe.py", line 117, in objective_fun_rec
    total += self.objective_fun_rec(params,N-1,str(i),p)
  File "nvqe.py", line 119, in objective_fun_rec
    total += self.objective_fun_rec(params,N-1,pauli_indexes+str(i),p)
  File "nvqe.py", line 110, in objective_fun_rec
    expected, neg = self.calculate_expected(result.get_counts(qc),pauli_indexes)
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/result/result.py", line 176, in get_counts
    exp = self._get_experiment(experiment)
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/result/result.py", line 270, in _get_experiment
    key)
qiskit.exceptions.QiskitError: 'Data for experiment "circuit15860" could not be found. '
```

Steps to reproduce the problem

<https://gist.github.com/shreychowdhary/beb0bc8c9ad2d2908536a2019b72073f>
while running using qiskit-aer-gpu

What is the expected behavior?

Its should run through without any errors which it does if I do not use the statevector_gpu backend option.

Suggested solutions

None



shreychowdhary added **bug** on Mar 15, 2020

shreychowdhary changed the title ~~Using GPU cause Data to not be found~~ Using GPU causes Data to not be found on Mar 15, 2020

atilag assigned [doichanj](#) on Mar 16, 2020

Assignees

[doichanj](#)

[hhorii](#)

Labels

bug

Type

No type

Projects

No projects

Milestone

Aer 0.5.1

Closed on Apr 28, 2020, 100% complete

Relationships

None yet

Development

[fix memory leak in gpu simulator](#)

Qiskit/qiskit-aer

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Participants

[shreychowdhary](#) [doichanj](#) [hhorii](#) [atilag](#) +1

 **shreychowdhary** added **bug** on Mar 15, 2020

 **shreychowdhary** changed the title **Using GPU cause Data to not be found** Using GPU causes Data to not be found on Mar 15, 2020

 **atilag** assigned **doichanj** on Mar 16, 2020

 **chriselectic** assigned **hhorii** on Apr 14, 2020

 **chriselectic** added this to the **Aer 0.5.1** milestone on Apr 15, 2020

 **jwoehr** on Apr 16, 2020 · edited by jwoehr

Edits ▾ Contributor ...

My code is using master branch (updated daily, it's 2020-04-15 today) I'm experiencing the same error.

- The BasicAer simulators return results.
- The Aer statevector gpu simulator and density matrix gpu simulator both return the error shown by [@shreychowdhary](#) due to no results.



 **chriselectic** on Apr 20, 2020

Member ...

[@jwoehr](#), are any GPU tests passing for you? (If you have the git repo checked out run `python -m unittest test.terra.backends.test_qasm_simulator_statevector.TestQasmSimulatorStatevectorThrustGPU`)

[@hhorii](#) [@doichanj](#) Have you been able to look into this yet?



 **jwoehr** on Apr 20, 2020 · edited by jwoehr

Edits ▾ Contributor ...

[@chriselectic](#) I have master branch checked out, yes, but can't get that command to work for me ... what directory should I be in to execute it?

```
$ python -m unittest test.terra.backends.test_qasm_simulator_statevector.TestQasmSimulatorStatevectorThrustGPU
E
=====
ERROR: terra (unittest.loader._FailedTest)
=====
ImportError: Failed to import test module: terra
Traceback (most recent call last):
  File "/usr/lib/python3.6/unittest/loader.py", line 153, in loadTestsFromName
    module = __import__(module_name)
ModuleNotFoundError: No module named 'test.terra'
```



 **jwoehr** on Apr 22, 2020

Contributor ...

[@hhorii](#) I got a notification from you about a similar error on the Mac but I can't find your message here ...



 **chriselectic** on Apr 22, 2020

Member ...

[@jwoehr](#) That command should be executed in the `qiskit-aer` git repo directory (you may need to have installed in executable mode `pip install -e .` for it to work properly)





Contributor ...



Collaborator ...

GPU simulator may have memory leak.

The status of execute error was ERROR: std::bad_alloc: cudaErrorMemoryAllocation: out of memory.



 hhorii mentioned this on Apr 26, 2020

[fix memory leak in gpu simulator #717](#)

Collaborator ...

@jwoehr thank you for your report. I was able to fix it. I believe this fix will be included in the next release (5.2).



Contributor ...

Thank you @hhorii I pull `master` branch daily, I'll watch for your pull request to be accepted!

  **atilag** closed this as **completed** in **#717** on Apr 27, 2020

Edits ▾ Contributor ...

@atilag @hhorii : I pulled the latest and I don't see any difference in behavior:

```
File ~/home/jax/work/QISKit/qiskit_dev_venv4/lib/python3.6/site-packages/qiskit_terra-0.14.0-py3.6-linux-x86_64-qiskit-terra.py:179:
    return self._get_experiment(experiment).data.to_dict()
File ~/home/jax/work/QISKit/qiskit_dev_venv4/lib/python3.6/site-packages/qiskit_terra-0.14.0-py3.6-linux-x86_64-qiskit-terra.py:179:
    key)
qiskit.exceptions.QiskitError: 'Data for experiment "circuit0" could not be found.'
```

Here's the job dictionary after the run:

```
{ 'args': (('method': 'density_matrix_gpu', None, False),
          'backend': <QasmSimulator('qasm_simulator') from AerProvider()>,
          'fn': <bound method AerBackend.run_job of <QasmSimulator('qasm_simulator') from AerProvider()>>,
          'future': <Future at 0x3760a90ee58 state:finished returned:Result>}
```

```
qiskit.exceptions.QiskitError: 'Data for experiment "circuit0" could not be found.'
```

Here's the job dictionary after the run:

```
{ '_args': ({'method': 'density_matrix_gpu'}, None, False),
  '_backend': <QasmSimulator('qasm_simulator') from AerProvider(>,>,
  '_fn': <bound method AerBackend.run_job of <QasmSimulator('qasm_simulator') from AerProvider(>,>,>,
  '_future': <Future at 0x7f69c9ccaa58 state=finished returned Result>,
  '_job_id': '117d78d5-e76c-47c1-8251-fd38452173c2',
  '_qobj': QasmQobj(qobj_id='9c1b6252-93bf-493c-9838-89035ca2dbb5', config=QasmQobjConfig(backend_options={
```



hhorii on Apr 28, 2020 · edited by hhorii

Edits Collaborator ...

@jwoehr Your first error was on `circuit15860`. This means that your 0-15859-th simulation were finished successfully and 15860-th simulation failed.

Your last error was `circuit0`. I believe that your simulator failed without any. Its reason may be that you did not build GPU simulator.

Could you tell me you specified `-DAER_THRUST_BACKEND=CUDA` or not to build the master?

Ex: `python setup.py bdist_wheel -- -DAER_THRUST_BACKEND=CUDA -- -j8`

I think you can find a hint of an error reason by checking `result` object)

```
result = execute(qc, backend, backend_options={"method": "statevector_gpu", "max_parallel_experiments": 1}
if (not result.success):
    print(result)
```



jwoehr on Apr 28, 2020

Contributor ...

@hhorii

Ex: `python setup.py bdist_wheel -- -DAER_THRUST_BACKEND=CUDA -- -j8`

Ha! Did not realize I had to do that :)

Where is the build process for Aer GPU documented so I can point the next person who asks me to this step?



atilag on May 21, 2020

Member ...

Ouch, I just find out about this message.

And you are right, the install process is documented but not the build one.

I'll document it!

